清华大学能源与动力工程系 2021 春《燃烧学》期中考试
（A 卷）
Middle Term Examination of Combustion (A)

考试时间(Date)： 2021 年 6 月 12 日 班级(Class)：_____

学号(Student No.)：_____ 姓名(Name)：_____

提示 (Instructions):

1 请将你的学生证放在桌子的右上角（Please put your student ID on the desk）

2 考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they cannot be lent to others. You should answer the questions independently and help is not allowed）

3 试卷共有**5**题，你要在**120**分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有**5**页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are **5** questions on this exam with **6** pages. You have up to **120** minutes to work on this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）

4 如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）

5 强烈推荐你先读一下所有的问题，并对回答问题的时间进行安排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）

6 请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸首页上标明你的试卷的类型（A或B卷）（**PRINT your name NOW** in the spaces at the top of your pages. Mark the exam type A or B）

7 答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

一、简答题：（42 分）

（6 分）1.定压球形膨胀火焰来测量火焰速度，且火焰速度满足$S_L = \frac{\rho_b}{\rho_u}\frac{dr_f}{dt}$，如果将初始燃料-空气混合物的温度从 300K 提高 600K，尝试判断火焰速度是增加还是减少？大体上会变化多少？

A constant-pressure spherical expanding flame is used to measure the burning velocity, and the burning velocity satisfies $S_L = \frac{\rho_b}{\rho_u}\frac{dr_f}{dt}$. If the temperature of initial fuel-air mixture is increased from 300K to 600K, please determine whether the flame speed increases or decreases? How much will it change in general?

（6 分）2. 请问守恒标量的定义是什么？请提出至少四个守恒标量。

What is the definition of a conserved scalar? Please suggest at least four conserved scalars.

（6 分）3. 液滴燃烧通常可以假设为双模模型，有人认为对于内区和外区，其产物的传质都符合 Stefan 流问题，您认为对吗？并解释理由。

The droplet combustion can usually be assumed to be a two-film model. Some think that the mass transfer of the product in the inner zone and the outer zone conforms to the Stefan problem. Do you think it is correct? And please explain.

（6 分）4. 什么时候可以认为液滴蒸发是准稳态的？

When can it be considered that droplet evaporation is quasi-steady?

（6 分）5. 请举例四种测量层流火焰速度的方法。

Please give examples of four methods for measuring laminar flame velocity.

（6 分）6. 在层流非预混火焰的建模中，出现了如下温度 T 与混合物分数 f 的线性关系：

$$T(f) = [f - Y_F(f)]\frac{\Delta h_c}{c_p} + f\left(T_{F,e} - T_{Ox,\infty}\right) + T_{Ox,\infty}$$

请说出上式严格成立所用到的简化假设（至少写 3 条）

The following linear relation between temperature T and mixture fraction f is encountered in modeling laminar non-premixed flames:

$$\boldsymbol{T(f) = [f - Y_F(f)]\frac{\Delta h_c}{c_p} + f\left(T_{F,e} - T_{Ox,\infty}\right) + T_{Ox,\infty}}$$

Please list at least three assumptions that makes the above relation holds rigorously.

（6 分）7. 请判断式$\boldsymbol{D^2(t) = D_0^2 - K_B t}$及$\boldsymbol{K_B = \frac{8\rho D}{\rho_C} ln\,(1 + B)}$是否只适用于扩散控制的燃烧，并说明理由

Please judge whether the formula $D^2(\boldsymbol{t}) = D_0^2 - K_B t$ and$K_B = \frac{8\rho D}{\rho_C} ln\,(1 + B)$ is only applicable to diffusion-controlled combustion, and explain the reasons

（15 分）二、直径 40 微米的碳颗粒在体积比为 3:7 的氧气-氮气混合气中燃烧，气压为 2atm。采用三色法测得碳颗粒表面温度为 2000K。假设碳表面混合气体的平均摩尔质量为 30 kg/kmol，用氮气中二氧化碳的扩散系数代表在实验中的多组分扩散系数。在 293K 时，氮气在二氧化碳中的扩散系数为$\boldsymbol{1.63 \times 10^{-5} m^2/s}$。试采用单膜模型求解此时颗粒表面 CO2 质量分数，其中表面 C-CO2 反应速率常数为$\boldsymbol{k_c = 4.016 \times 10^8 \exp\left(\frac{-29790}{T_s(K)}\right) m/s}$。（两次迭代的相对误差小于 5%即可）。

Carbon particles with a diameter of 40 microns are burned in an oxygen-nitrogen mixture with a volume ratio of 3:7 at a pressure of 2 atm. The surface temperature of the carbon particles measured by the three-color method is 2000K. Assuming that the average molar mass of the mixed gas on the carbon surface is 30 kg/kmol, the diffusion coefficient of carbon dioxide in nitrogen is used to represent the multi-component diffusion coefficient in the experiment. At 293K, the diffusion coefficient of nitrogen in carbon dioxide is $1.63 \times 10^{-5} m^2/s$. Try to use the single film model to solve the CO_2 mass fraction on the particle surface at this time, where the surface C-CO2 reaction rate constant is $k_c = 4.016 \times 10^8 \exp\left(\frac{-29790}{T_s(K)}\right) m/s$. (The relative error of the two iterations is less than 5%).

(14 分)三、层状火焰速度是可燃混合物最重要的内在性质之一。由于其重要性，存在多种方法来测量层状火焰速度。其中一种叫做球形火焰测量方法，又分为定容球形火焰和定压球形火焰。

1.在一无限大空间中充满了某气体燃料和空气的混合物，如果在某点点火，可以测得火焰面传播速度 S，请据此推导气体速度在 t 时刻的分布。

2.若将无限大空间换为半径为 R 的球壳，其中充满了某气体燃料和空气的混合物，如果在球心点火，可以测得容器内部压力的变化曲线 P(t)，请据此推导气体速度在 t 时刻的分布。

（推导过程需要的变量和条件请自行给出）

The lamellar flame velocity is one of the most important intrinsic properties of combustible mixtures. Due to its importance, there are multiple methods to measure laminar flame velocity. One of them is called spherical flame measurement method, which is divided into constant volume spherical flame and constant pressure spherical flame.

1. An infinite space is filled with a mixture of gaseous fuel and air. If it is ignited at a certain point, the flame surface propagation velocity S can be measured. Please derive the gas velocity distribution at time t based on this.

2. If the infinite space is replaced by a spherical shell with a radius of R, which is filled with a mixture of a certain gas fuel and air, if the ignition is in the center of the sphere, the change curve P(t) of the internal pressure of the container can be measured, please follow this Derive the distribution of gas velocity at time t.

(Please give out the variables and conditions needed in the derivation process)

（15 分）四、考虑一个正辛烷（C_8H_{18} ）和乙醇(C_2H_5OH)燃料液滴的燃烧，其直径为 200 μm。假设，环境静止，液滴温度等于其沸点，当$P = 1\ atm$ 及$T_\infty = 300\ K$时，分别求：

(1) 质量燃烧速率；

(2) 火焰半径的比较

	正辛烷	乙醇
沸点 K	398.75	351.55
摩尔质量 kg/kmol	114.22	46.07
产物的平均摩尔质量 kg/kmol	28.6	28.12
液滴密度 kg/m^3	703	789
汽化潜热 kJ/kg	302	820
燃烧热值 kJ/kg	47890	26900
燃料蒸气比热容 kJ/(kg·K)	5.1	5.1
空气比热容 kJ/(kg·K)	2.2	
1000K 下空气热导率 W/(m·K)	0.067	
1000K 下燃料气体热导率 W/(m·K)	0.097	
空气摩尔质量 kg/kmol	28.85	

Consider the combustion of a droplet of n-octane (C_8H_{18}) and ethanol (C_2H_5OH) fuel with a diameter of 200 m. Assuming that the environment is static and the droplet temperature is equal to its boiling point, when P=1 atm and T_∞ =300 K, respectively find:

(1)Mass burning rate;

(2)Comparison of flame radius

	N-octane	ethanol
Boiling point K	398.75	351.55
Molar mass kg/kmol	114.22	46.07
The average molar mass of the product kg/kmol	28.6	28.12
Droplet density kg/m3	703	789
Latent heat of vaporization kJ/kg	302	820
Combustion heat value kJ/kg	47890	26900
Specific heat capacity of fuel vapor kJ/(kg·K)	5.1	5.1
Specific heat capacity of air kJ/(kg·K)	2.2	
Air thermal conductivity at 1000K W/(m·K)	0.067	
Thermal conductivity of fuel gas at 1000K W/(m·K)	0.097	
Air molar mass kg/kmol	28.85	

（14 分）五、氨裂解（$NH_3 \rightarrow N_2 + H_2$）是工业制氢的方式之一，这是一个吸热反应，1mol 的氨裂解需吸收 46.22 kJ 的热量。某公司生产的氨裂解炉用电加热，参数如下：

产氢速率	**11.25 m^3/h (标准状态，即 273K, 1atm)**
耗电功率	**15 kW**

请计算：

（1） 该系统热效率；

（2） 将该裂解炉重新设计，氢的产率不变，所需热量全部来自氨燃烧（产物是 N_2 和水）。假设系统热效率不变，若燃烧器采用方形喷口，符合层流非预混火焰，一次风率 13.8 %，则需要多少个喷口，才能使各火焰长度不大于 100 mm？假设各喷口火焰独立，氨的热值取为 18.6 MJ/kg，环境及燃料流入射温度均为 300 K。

Ammonia cracking ($NH_3 \rightarrow N_2 + H_2$) is an industrial process for hydrogen production. It is an endothermic reaction, and the cracking of 1mol ammonia requires 46.22 kJ energy input. A company produces electrically-heated ammonia cracking furnaces, with parameters listed as below:

Hydrogen capacity	11.25 m^3/h (Standard state, that is, 273K & 1atm)
Power consumption	15 kW

(i) What is the thermal efficiency of the furnace?

(ii) Redesign the furnace to be heated solely by ammonia combustion (which forms N_2 and H_2O). The capacity of hydrogen keeps the same. Assume that the thermal efficiency is unchanged. All the burners are featured with square ports and laminar non-premixed flames. Assume a 13.8 percent primary aeration. Then how many ports are required so that the flame length of each port is no larger than 100 mm? Assume all the flames are independent. Take the heat value of ammonia as 18.6 MJ/kg. The ambience and fuel jet temperatures are both 300 K.